CMD: Continuous Morphological Dynamics — Sensory Grounding and Thermodynamic Habituation

Irfan Mahir — Independent Researcher, Dhaka, Bangladesh — irfan@furylogic.com

Contents

CMD: Continuous Morphological Dynamics — Sensory Grounding and Thermodynamic Habituation **1**
Abstract . . . 1
1. Why a Sensory Layer Is Necessary . . . 1
2. Byte-Lane Grounding . . . 2
2.1 Structure . . . 2
2.2 Binding by rarity . . . 2
2.3 Consolidation and starvation . . . 2
2.4 Why recognition is graded . . . 2
2.5 A representational limit, and the escape . . . 3
3. Thermodynamic Habituation . . . 3
3.1 The stopword problem, without a stopword list . . . 3
3.2 The Curie gate . . . 3
3.3 What this buys . . . 3
4. Wavefront-Gated Emission . . . 4
5. Integration . . . 4
6. The Failure This Layer Fixed . . . 4
7. What CMD Does Not Solve . . . 5
8. Conclusion . . . 5
References . . . 5

CMD: Continuous Morphological Dynamics — Sensory Grounding and Thermodynamic Habituation

Irfan Mahir *Independent Researcher, Dhaka, Bangladesh* irfan@furylogic.com

> *"A stopword is not a word on a list. It is a cavity that has heated itself past its Curie point."*

Abstract

This paper specifies the substrate layer that later work in this lineage cites as *CMD* and, until now, could not point at. Three mechanisms are defined.

Byte-lane sensory grounding. Concepts are not reachable by name. Input bytes drive 256 dedicated sensory lanes, and energy reaches a concept only through *learned bindings* from those lanes. Recognition therefore involves no string comparison and is **graded**: a misspelling still drives most of the correct lanes and still resonates the right concept, at reduced amplitude.

Thermodynamic habituation. Each cavity carries a heat variable integrating its own activity, $\dot{T} = \alpha|v| - \beta T$. Tunnel coupling is gated by $e^{-T/T_{\text{crit}}}$, so a concept that fires on every input **decouples itself** from the graph. This replaces the stopword list with a physical process: ubiquity is punished by the dynamics rather than by a curated exclusion set.

Wavefront-gated emission. Readout is restricted to the region the input actually reached, so a settled field does not report cavities that were merely nearby.

Section 6 reports the failure this layer was built to fix — a substrate that answered `minecraft` with `alexandria` — and §7 states plainly which half of the emission problem CMD does **not** solve.

---

1. Why a Sensory Layer Is Necessary

The obvious way to inject a query into a concept substrate is to look up the concept and drive its cavity. This has two defects, and the second is fatal.

It is brittle. Lookup is a hash probe: exact or nothing. One wrong character and the concept does not exist, which is not how recognition works in anything that recognises.

It is ungrounded. If concepts are reachable by name, the "physics" is decoration on a dictionary — the retrieval already happened, in the hash table, before any wave propagated. A substrate whose entry point is a lookup has conceded the interesting part.

CMD's response is to make the *bytes* the sensory surface and require that everything above be reached through learned coupling.

---

2. Byte-Lane Grounding

2.1 Structure

The substrate carries 256 **byte lanes**, one per possible input byte, existing alongside the concept cavities. A lane is an oscillator like any other; what distinguishes it is that it is the only thing external input can touch directly.

Each lane holds a sparse map of **binding weights** to concept cavities:

```
byte_bind_w[b] : cavity_id -> weight
```

Bindings **start at zero**. A byte reaches a concept only insofar as a binding has been established, and until then the input goes nowhere. This is the property that makes the layer sensory rather than addressing.

2.2 Binding by rarity

Bindings are established at grounding time with weight equal to inverse lane document-frequency:

$$w_{b,c} \;=\; \log \frac{N}{1 + \mathrm{df}(b)}$$

for N concepts and $\mathrm{df}(b)$ the number of labels containing byte b. A byte occurring in nearly every label distinguishes nothing and binds weakly; a rare byte binds strongly.

This is the same principle as the Curie gate of §3 applied at grounding time instead of during propagation: **ubiquitous participants are suppressed by construction, not by a rule.**

2.3 Consolidation and starvation

Bindings are not static. They grow under co-activity and decay under disuse:

$$w \leftarrow \tanh\left(\left(w\,\lambda_{\text{decay}} + \eta\, v_b\, v_c\right) / w_{\max}\right) w_{\max}$$

with growth rate $\eta = 0.02$, decay $\lambda = 0.9995$ per tick, and a saturating ceiling $w_{\max} = 0.6$. The decay term is what allows the sensory layer to forget a binding that stops paying — starvation rather than deletion.

2.4 Why recognition is graded

Because a concept accumulates energy from *all* the lanes its label touches, a query that shares most of its bytes with a label still delivers most of the energy. `metfromin` drives nearly the same lane set as `metformin` and resonates the same concept at slightly lower amplitude. A hash lookup returns nothing for the same input.

This is the falsifiable consequence of the design, and it is the right thing to test the layer against: if recognition were not graded, the sensory layer would be an expensive hash table.

2.5 A representational limit, and the escape

Single-byte lanes cannot distinguish anagrams. `metformin` and `metfromin` contain the same byte multiset, and a multiset is invariant under permutation, so **no reweighting of single-byte lanes can separate them.** This is a theorem about the representation, not a tuning failure, and it was mistaken for a tuning failure first.

The escape is to change what is represented: **ordered pairs**. Lanes indexed by bigrams, bracketed with begin- and end-of-word sentinels so that word-initial and word-final position are themselves observable, distinguish the two words because the pair multisets differ. The theorem about byte multisets remains true; the sensory layer simply stopped being one.

---

3. Thermodynamic Habituation

3.1 The stopword problem, without a stopword list

Some concepts participate in almost everything. In a spreading-activation substrate they accumulate energy from every direction and dominate every readout. The conventional fix is a curated exclusion list, which is brittle, domain-specific, and hand-maintained.

CMD instead gives every cavity a **heat** variable integrating its own recent activity:

$$\frac{dT}{dt} \;=\; \alpha\,|v| \;-\; \beta\,T$$

Activity heats a cavity; inactivity cools it. The equilibrium temperature of a cavity is proportional to how often it moves.

3.2 The Curie gate

Tunnel coupling is scaled by the heat of **both** endpoints:

$$K_{\text{eff}} \;=\; K_0\, e^{-T_a/T_{\text{crit}}}\, e^{-T_b/T_{\text{crit}}}$$

with $T_{\text{crit}} = 3.0$. Below the critical temperature the gate is near unity and the tunnel transmits normally. Above it, transmission collapses.

A concept that fires on every input therefore cooks itself out of the field. Rare concepts stay cold and consolidate strong bonds. The analogy is exact rather than decorative: above its Curie point a ferromagnet loses its ordering, and above T_{crit} a cavity loses its coupling.

3.3 What this buys

The mechanism is **automatic, continuous, and domain-independent**. It requires no list, adapts as the ontology changes, and degrades smoothly rather than at a boundary. A concept that is ubiquitous in one corpus and rare in another is suppressed in the first and not the second, without configuration.

It also composes: because heat is a per-cavity scalar read by other mechanisms, an external verdict can raise it. A hypothesis refuted by experiment drives its target cavity, which heats it, which decouples it — so **unreliability becomes a physical property of a concept** rather than an annotation about it.

---

4. Wavefront-Gated Emission

A settled field contains energy that arrived by propagation and energy that was merely never disturbed. Reading out by amplitude alone conflates them.

CMD tracks, per cavity, how recently the wavefront touched it, and restricts readout to cavities the input actually reached. The gate is a continuous decay rather than a flag, so recency is graded like everything else in the layer.

The practical consequence is that a query about one domain does not report high-amplitude cavities from another simply because they were already vibrating.

---

5. Integration

CMD provides the state that later mechanisms read. Each consumes it differently, and none duplicates it:

Mechanism	Reads	Uses it for
Curie gate (this paper)	T_i	collapsing coupling of ubiquitous cavities
Chaotic metastability	T_i	shaking *entrenched* cavities hardest, via $\tanh(T_i/T_{\text{crit}})$
Phase-consistent Hebbian	binding weights	consolidating only phase-consistent co-activity
Momentum emission	wavefront recency	gating which cavities may be emitted

The chaos coupling is worth noting: it and the Curie gate read the *same* variable to opposite ends. Heat makes a cavity decouple, and heat also makes it the one most strongly perturbed. Entrenchment is thereby both punished and disrupted, which is what allows escape rather than mere silencing.

---

6. The Failure This Layer Fixed

CMD was not designed in the abstract. It replaced a mechanism that demonstrably did not work.

The prior substrate injected input bytes into cavities selected by a **hash of the byte value modulo the cavity count**. This is arbitrary addressing wearing the costume of sensation: byte `m` would land on whatever cavity index the hash produced, with no relationship between the byte and the concept.

The observable failure was that a query for `minecraft` returned `alexandria`. The two share bytes; the hash scattered those bytes into unrelated cavities; energy accumulated somewhere structurally meaningless; and the readout reported it confidently.

The fix has three parts, and all three are necessary. A **dedicated lane layer** so bytes have somewhere of their own to go. **Bindings starting at zero** so a byte cannot reach a concept it has never been associated with. And **rarity weighting** so that common bytes cannot carry a match on their own.

The general lesson is stated in the successor literature as the difference between a tuning problem and a representational one: no adjustment to the hash could have fixed this, because the hash was not approximately right.

---

7. What CMD Does Not Solve

Emission of novel byte sequences. CMD grounds *recognition* in bytes. The symmetric operation — originating a byte sequence for a word not in the lexicon — does not work. Sweeping the byte-lane phase field to emit characters babbles.

The asymmetry is structural rather than incidental. **Recognition is a comparison problem** and yields to better representation, which is why the bigram widening of §2.5 fixed it. **Emission is a generation problem**: the field must *originate* a sequence, with nothing to compare against. No amount of representational widening supplies the missing mechanism, and widening the alphabet — the fix that worked for recognition — does not help.

This is the residual of the emission problem, and the honest scope of this paper is that it solves the input half.

Heat is not relevance. A cold cavity is one that has not fired recently, which is not the same as one that matters. Habituation suppresses the ubiquitous; it does not promote the important.

Constants are set, not derived. $T_{\text{crit}} = 3.0$, α, β, and the binding rates are chosen values. The material constants elsewhere in this substrate were replaced by a derivation from relation fan-out and measurably improved; the thermal constants have not had the same treatment, and until they do this layer carries the same unfalsifiability the material table used to.

---

8. Conclusion

CMD makes the entry point to the substrate sensory rather than addressed. Concepts become unreachable by name and reachable only through learned coupling from byte lanes, which makes recognition graded, makes misspelling a matter of degree, and removes the hash table that was doing the retrieval before any physics ran.

Habituation then replaces a curated stopword list with a thermodynamic process, and does so in a form other mechanisms can read — which is why a refuted hypothesis can heat a concept and

a chaotic driver can shake the hottest one.

The layer is honest about its half of the problem. Input is grounded; output is not.

References

- Mahir, I. *PRM: The Phonic Resonance Matrix.* — the cavity and tunnel substrate CMD grounds.
- Mahir, I. *MIMS: Mechanical Impedance and Material Stiffness.* — the material derivation §7 contrasts with CMD's set constants.
- Mahir, I. *MRS: Morphological Resonance Synthesis.* — the bigram extension of §2.5, and the emission half that does not work.
- Mahir, I. *CSAT: Chaotic Strange-Attractor Trajectory.* — reads T_i to opposite purpose.
- Mahir, I. *CPME / CPME-II.* — momentum emission and phase-consistent consolidation built on this layer.
- Mahir, I. *From Artificial to Synthetic Intelligence.* §5.4c, §8.5.
- Implementation: `src/physics.hpp` (`byte_bind_w`, `bind_bigram`, `heat_T`, `T_crit`, `drive_byte`, wavefront gating).